Reproducibility Checklist

This checklist is filled for the current BGR submission package. The main paper is anonymous; commands, configs, per-seed CSVs, and run ledgers are in the anonymous artifact's `README.md` and `results/README.md`.

1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA)

yes. The method section defines replayable states and recovery curves. It also states critical radii, BGR priority, and a training-loop method box.

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no)

yes. The paper separates controlled margin-expansion claims, robot-suffix simulator results, OpenVLA fixed-policy audits, OpenVLA-OFT adaptation diagnostics, and tabular grid-policy setup failures.

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no)

yes. The related-work section cites curriculum learning, PLR, PAIRED/ACCEL, domain randomization, MiniGrid, LIBERO, and OpenVLA.

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no)

yes. The paper introduces a recovery-margin formalism and states a local margin-update proposition for boundary-centered radius sampling.

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no)

yes. Proposition 1 states the local update-kernel and stochastic-dominance assumptions; the limitations section states replay-access and perturbation-family restrictions.

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no)

yes. The local theoretical claim is stated as Proposition 1. Empirical claims are stated with measured metrics and paired-seed tests.

2.4. Proofs of all novel claims are included (yes/partial/no)

yes. Proposition 1 includes a proof sketch based on monotone coupling and monotonicity of the update kernel.

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no)

yes. The method section gives the local margin-update intuition and the proposition proof sketch.

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no)

yes. The proof cites a standard stochastic-orders reference for the quantile coupling used in Proposition 1.

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA)

partial. The proposition is a local model; the grid ablation empirically tests its main implication that boundary-centered radius sampling matters.

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA)

yes. Ablation scripts, configs, per-seed CSVs, and claim-checking scripts are included.

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no)

partial. Main BGR training results use procedural synthetic/grid/suffix generators. The LIBERO/OpenVLA recovery audit uses existing benchmark task definitions, init states, and fixed-policy rollout artifacts to validate learned-policy recovery-curve measurement.

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA)

yes. The paper motivates synthetic recovery curves, procedural grid-margin replay states, robot suffix diagnostics, and LIBERO/OpenVLA as a realistic manipulation-policy target.

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA)

yes. The experiments are generated by artifact-provided scripts/configs; per-seed result CSVs are included under `results/`.

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA)

yes. The generated procedural data and code are included in the anonymous artifact under an MIT license.

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA)

yes. LIBERO and OpenVLA are cited; MiniGrid is cited for grid-style context.

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA)

yes for LIBERO task definitions, benchmark assets, and OpenVLA model code/weights. The anonymous artifact contains compact derived recovery summaries; raw fixed-policy rollout logs are retained for audit but are not required for the claimed quantitative results.

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA)

NA. No non-public dataset is used for claimed quantitative results.

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no)

yes.

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA)

partial. Final configs and ablation variants are included in the anonymous artifact; the full search history is not exhaustively enumerated in the main paper.

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no)

yes. Procedural generation, LIBERO probing, OpenVLA recovery summarization, aggregation, and figure scripts are included.

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no)

yes. Experiment scripts, configs, tests, aggregation, and result CSVs are included in the anonymous artifact.

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no)

yes. The anonymous artifact includes an MIT license, and the code plus generated procedural data will be made publicly available upon publication.

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no)

partial. Code is modular and tested; comments are intentionally sparse and do not yet cross-reference paper sections.

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA)

yes. Every config lists seeds; scripts pass seeds to NumPy generators and result CSVs include per-seed rows.

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no)

yes. The run ledger records Slurm partitions, CPU/memory/time requests, hostnames for logged runs, environment constraints, and JSON environment snapshots for compute and GPU jobs.

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no)

yes. Recovery AUC, critical radius, clean success, transfer RAUC, and AULC are defined or described.

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no)

yes. Runs are reported in the paper and run ledger: synthetic training, estimator validation, grid-margin full-baseline, grid ablation, grid sensitivity confirmations, and robot-suffix coverage comparisons use 30 paired seeds; grid learning-curve diagnostics use 15 seed pairs. Smaller three- or five-seed exploratory diagnostics are labeled as diagnostics in the run ledger rather than used as main claims.

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no)

yes. Aggregated tables include standard errors; experiments report median critical radius and AULC in addition to final means.

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no)

yes. The anonymous artifact includes paired exact sign tests for the reported paired comparisons, including the 30-seed suffix coverage confirmation, grid-margin target-sensitivity, grid-margin obstacle-regime sensitivity, grid-margin geometry-stress sensitivity, and grid-margin learning-curve sweeps. The paper separates these main comparisons from smaller exploratory evidence.

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA)

yes. Final hyperparameters are in the artifact YAML configs under `configs/`.